War Carbon Emissions Dashboard

Methodology v1.0.5

Working Draft — Subject to Revision

WCED Project

Last updated: 2026-05-24

This document specifies the current emission estimation methodology used by the War Carbon Emissions Dashboard. Equations, parameters, and decision rules are subject to revision; each revision increments the methodology version and is documented in the accompanying changelog (Section 6). An academic preprint with motivation, related work, and results is in preparation.

1 Scope

WCED v1.0.5 quantifies CO_2 released by hydrocarbon combustion at oil and fuel infrastructure in the Iran–Gulf theatre of the 2026 Iran–Israel–US war. This document specifies the equations, parameter priors, verification rules, and reconciliation logic used by the estimation system.

Version 1 covers refineries, crude and product depots, and petrochemical complexes from 2026-02-28 onward. It does *not* cover destroyed-building embodied carbon, munitions embodied carbon, combat aviation fuel, shipping rerouting, reconstruction, or non-carbon ecological damage; each of those is deferred to a later version. Emission estimates represent CO_2 released from observed fires and are not attributed to specific belligerents.

2 Estimation Framework

All emission estimates are expressed as probability distributions characterised by a 5th percentile (p5), median (p50), and 95th percentile (p95), computed by Monte Carlo simulation. Every result carries a provenance record tracing it through a chain of inputs back to primary satellite observations; no estimate is published as a scalar without bounds.

Each fire event is assigned a confidence label that reflects the evidence supporting it. Labels are defined in Section 4 and propagate forward into every downstream emission estimate, so the uncertainty associated with event identification is never decoupled from the reported emission range.

The two core emission methods—Fire Radiative Power (FRP)-based (Section 3.1) and inventory-based (Section 3.2)—are computed independently and then reconciled (Section 3.3). All emission factors and parameter prior distributions are maintained in versioned reference tables external to the estimation logic; they are not hard-coded.

3 Emission Calculations

Sections 3.1–3.2 (reserved)

Facility registry schema and emission factor tables are maintained as versioned reference data and are not reproduced here.

3.1 FRP-based Method

Fire Radiative Power (FRP) observations from VIIRS, MODIS, and Sentinel-3 SLSTR are clustered into an event-level time series from which a total fire radiative energy is derived. The method proceeds in four stages: raw energy integration, baseline subtraction, overpass-gap correction, and conversion to CO_2 mass.

Stage 1: Raw energy integration. The pipeline begins from satellite FRP observations sampled at overpass times. The raw Fire Radiative Energy is the trapezoidal integral over the observation window:

$$I_{\mathrm{raw}} = \int_{t_0}^{t_1} \mathrm{FRP}(t)\,\mathrm{d}t. \tag{1}$$

Stage 2: Pre-war baseline subtraction. The raw integral over-counts because operating facilities flare routinely. A facility-specific pre-war baseline is subtracted to isolate the conflict-attributable signal. The baseline is derived from 12 months of archival satellite observations preceding the conflict (2025-02-28 to 2026-02-27).

For each facility, the rolling 30-day pre-war baseline FRP and its uncertainty are:

$$\mathrm{FRP}_{\mathrm{baseline}} = P_{75}\big(\{\mathrm{FRP}_i\}_{i \in W_{30}}\big), \tag{2}$$

$$\sigma_{\mathrm{baseline}} = \mathrm{IQR}\big(\{\mathrm{FRP}_i\}\big)/1.349, \tag{3}$$

where W_{30} is the 30-day window preceding the event, P_{75} denotes the 75th percentile (capturing the upper envelope of routine operations), and the IQR-based estimator provides a robust standard deviation resistant to transient flare-ups. The net energy available for conflict attribution is:

$$I_{\mathrm{net}} = \max\big(0,\ I_{\mathrm{raw}} - \mathrm{FRP}_{\mathrm{baseline}} \cdot \Delta t\big), \tag{4}$$

where Δt is the event duration in days. Events for which the 30-day window contains fewer than five observations use a conservative fallback baseline ($\mu = 0$ MW, $\sigma = 50$ MW) and are flagged as having insufficient baseline history.

Stage 3: Overpass-gap correction. Satellites sample at discrete overpass times. Two multiplicative corrections account for fire activity in unobserved intervals and for residual sampling bias. The corrected Fire Radiative Energy is:

$$I_{\mathrm{FRE}} = k_{\mathrm{ext}} \cdot d \cdot I_{\mathrm{net}}, \tag{5}$$

where d is the burn duty cycle, representing the fraction of time the fire is active between overpasses (prior: triangular(0.40, 0.70, 0.95); informed by the Kuwait 1991 and Abqaiq 2019 oil-facility fires), and k_{ext} is an extrapolation factor correcting residual bias when overpass sampling is sparse (prior: $\mathcal{N}(1.0, 0.15)$, calibrated against MODIS/VIIRS cross-comparisons). Both parameters are sampled independently in the Monte Carlo (Section 5); applying either as a point value before sampling would systematically narrow the resulting uncertainty interval. I_{net} and I_{FRE} both carry units of MJ.

Stage 4: Conversion to CO_2 mass. Corrected energy converts to fuel mass through a calibrated coefficient, then to CO_2 mass through combustion stoichiometry:

$$m_{\text{fuel}} = \alpha \cdot I_{\text{FRE}}, \tag{6}$$
$$m_{\text{CO}_2} = m_{\text{fuel}} \cdot \beta \cdot r, \tag{7}$$

where α is the FRP-to-combustion-rate coefficient ($\mathcal{N}(0.368, 0.05)$ kg/MJ; Wooster et al. [3]), $\beta = (44/12) \cdot f_C$ is the stoichiometric $\text{C} \to \text{CO}_2$ conversion factor with $f_C \approx 0.86$ the carbon mass fraction of crude oil (EPA AP-42; [1]), and r is the carbon recovery fraction (triangular(0.92, 0.96, 0.98); Hobbs & Radke [4]). At central prior values, $\beta \cdot r \approx 3.15$ kg CO_2 per kg crude burned, consistent with the value used in the CCI Iran brief [11, 5].

3.2 Inventory-based Method

For facilities with a curated nameplate capacity C (barrels), a bottom-up estimate is computed from:

$$m_{\text{CO}_2} \;=\; C \cdot \phi \cdot \psi \cdot \text{EF}, \tag{8}$$

where ϕ is the fraction of nameplate capacity present at the time of the strike (prior: uniform(0.30, 0.90); a judgment-based prior pending empirical calibration), ψ is the destroyed fraction, and EF is the combustion emission factor for the relevant product:

- Crude depots and refinery crude units: EF = 0.425 tCO_2/barrel (EPA AP-42 §1.3; consistent with CCI Iran brief [5]).
- Refined-product depots: EF = 0.430 tCO_2/barrel (IPCC 2006 Guidelines, Vol. 2, Ch. 1, Table 1.3; [2]).

Each emission factor is designated applicable to specific facility types only; type-factor mismatches are rejected at the estimation stage. Applying a crude-oil factor to a gas-processing facility would produce systematically incorrect results and is therefore prohibited.

Facility-type restriction. The inventory method applies only to storage-type facilities (oil depots, storage tank farms, and refineries) where nameplate capacity represents stored inventory. It does not apply to production or processing facilities (gas processing plants, petrochemical complexes) where capacity represents throughput and on-site inventory is a small fraction of stated capacity. Events at production facilities use the FRP-based method exclusively.

Destroyed-fraction prior (v1.0.5 recalibration). The parameter ψ is event-specific: when a reviewer provides an explicit assessment $\hat{\psi}$ from before/after satellite imagery, the prior centres on that value using triangular($\hat{\psi} - 0.15,\ \hat{\psi},\ \hat{\psi} + 0.15$), truncated to $[0, 1]$. When no assessment is available, the v1.0.5 defaults in Table 1 apply.

The defaults were recalibrated in v1.0.5 after systematic FRP/inventory agreement ratios of 2.7–7.4× were observed under earlier defaults, indicating that initial priors substantially over-estimated the fraction of storage capacity destroyed in a typical strike:

The recalibrated mode of 0.15 reflects that strikes typically produce localised tank fires rather than facility-wide destruction, consistent with post-strike optical satellite imagery.

Table 1: Fraction-destroyed default priors by facility type and methodology version.

Facility type	Prior v1.0–v1.0.4	Prior v1.0.5	Basis
Oil depot	tri(0.20, 0.40, 0.65)	tri(0.05, 0.15, 0.30)	FRP/inv ratio calibration
Storage tank farm	tri(0.20, 0.40, 0.65)	tri(0.05, 0.15, 0.30)	Same
Refinery	tri(0.10, 0.20, 0.35)	tri(0.05, 0.15, 0.30)	Same

3.3 Reconciliation

Where both methods are applicable, they yield independent Monte Carlo distributions. Reconciliation follows three rules based on the ratio $\rho = m^{\mathrm{inv,p50}}_{\mathrm{CO_2}} \,/\, m^{\mathrm{frp,p50}}_{\mathrm{CO_2}}$:

Agreement ($0.5 \leq \rho \leq 2.0$): both estimates are published and the headline emission is the envelope distribution, formed by pooling the Monte Carlo samples from both methods ($2N = 20{,}000$ draws) and computing p5/p50/p95 over the combined set. This approach captures both within-method parametric uncertainty and between-method structural uncertainty in a single distribution, without averaging away genuine disagreement.

Disagreement ($\rho < 0.5$ or $\rho > 2.0$): both estimates are retained but neither is published as a headline figure until editorial review resolves the discrepancy. A written reviewer note must be entered in the public changelog before publication.

Third-party cross-check: emission estimates from external sources (e.g. CEOBS, CCI, or government statements) are recorded as reference values only and do not enter headline arithmetic.

Events where ρ falls within 10% of either threshold ($\rho \in [0.50, 0.55] \cup [1.82, 2.00]$) are flagged for additional editorial scrutiny even though they formally satisfy the agreement criterion.

4 Verification and Confidence Assignment

Each candidate fire event is assessed against three independent evidence streams:

- **Satellite persistence**: whether two or more independent satellite overpasses detect a hotspot at the same location.
- **Optical confirmation**: whether a cloud-free Sentinel-2 scene, acquired within ±72 hours of the detected event, confirms active combustion.
- **Conflict corroboration**: whether a documented armed event from a curated conflict database falls within a 24-hour, 2-kilometre spatio-temporal window of the detected fire.

Table 2 defines the resulting confidence labels.

The persistence threshold is two or more overpasses with elevated thermal signal. Conflict-record corroboration alone is intentionally non-promoting: a reported armed event near a facility without satellite confirmation may represent a near-miss, a non-incendiary strike, or a positional error in the conflict record. Such candidates are withheld from publication pending review.

Only CONFIRMED and VERIFIED events contribute to the headline emission total. REPORTED events appear on the public map with a distinct marker. SUSPECTED and CLAIMED events are withheld pending review.

Table 2: Confidence label decision table (v1.0.5).

Label	Persistent	Optical	Conflict	Notes
CONFIRMED	yes	yes	yes	All three streams agree
VERIFIED	yes	yes	no	Satellite-only; no conflict record
REPORTED	yes	no	any	Cloud cover or ambiguous scene
SUSPECTED	no	—	any	Conflict record alone never auto-promotes
SUSPECTED	no	—	no	Single overpass only
CLAIMED	—	—	—	News or official statement only; no satellite evidence

5 Uncertainty Quantification

Every emission estimate is the output of a Monte Carlo simulation with $N = 10{,}000$ independent draws. Parameter prior distributions are specified in Table 3; they are not hard-coded in the estimation logic and can be updated independently of it.

Table 3: Monte Carlo parameter priors.

Parameter	Prior distribution	Units	Source
Crude oil combustion factor	triangular(0.405, 0.425, 0.445)	tCO_2/barrel	EPA AP-42 §1.3 [1]
Refined product combustion factor	triangular(0.410, 0.430, 0.455)	tCO_2/barrel	IPCC 2006 GL Vol. 2 [2
FRP-to-combustion rate, α	$\mathcal{N}(0.368,\ 0.05)$	kg/MJ	Wooster et al. [3]
Carbon recovery, r	triangular(0.92, 0.96, 0.98)	—	Hobbs & Radke [4]
Burn duty cycle, d	triangular(0.40, 0.70, 0.95)	—	Expert prior (Kuwait 19
Facility inventory at strike, ϕ	uniform(0.30, 0.90)	—	Judgment-based prior
FRP extrapolation factor, k_{ext}	$\mathcal{N}(1.0,\ 0.15)$	—	MODIS/VIIRS cross-co
Fraction destroyed, ψ (storage; default)	triangular(0.05, 0.15, 0.30)	—	FRP/inventory ratio ca

Distribution semantics. Triangular distributions are parameterised as (low, mode, high). Where asymmetric bounds are not explicitly specified, symmetric bounds around the central value are assumed. Normal distributions use the stated standard deviation; reported uncertainty bounds in the source literature are stored for reviewer cross-check but are not used as truncation bounds.

Independence assumptions. Parameters are sampled independently for each event. Correlation of parameters across facilities—for example, systematic satellite calibration drift affecting all events simultaneously—is a planned extension for a later version.

Aggregation. Facility-level and theatre-level aggregates are computed by element-wise summation of per-event Monte Carlo sample arrays, then summarised to p5/p50/p95. This approach correctly propagates uncertainty through the summation rather than summing individual percentiles, which would understate aggregate uncertainty by assuming all events simultaneously attain their extreme values.

6 Methodology Version History

Version	Date	Type	Param. change	Summary
v1.0	2026-03-01	Initial	yes	Raw FRP method, inventory method, reconciliation, and confidence labels established.
v1.0.1	2026-05-24	Calibration	yes	Baseline subtraction introduced: 75th-percentile rolling baseline with IQR-based robust standard deviation. Net FRP computed prior to Monte Carlo sampling. Conservative fallback baseline for events with insufficient pre-war observations.
v1.0.2	2026-05-24	Data	no	Twelve months of pre-war archival satellite observations ingested. No equation or parameter change; baseline window now has sufficient historical coverage for all tracked facilities.
v1.0.3	2026-05-24	Recompute	no	Full recomputation under v1.0.2 baselines. FRP/inventory agreement ratios of 2.7–7.4× observed for storage-type facilities, indicating systematic overestimation of fraction destroyed. No parameter change.
v1.0.4	2026-05-24	Recompute	no	Confirmed systematic overestimation with additional data. No parameter change.
v1.0.5	2026-05-24	Calibration	yes	Fraction-destroyed defaults for storage-type facilities recalibrated to triangular(0.05, 0.15, 0.30). Damage assessment records updated to reflect recalibrated defaults.

This document supersedes methodology v1.0 (2026-05-23). All equation numbers are unchanged relative to v1.0; material changes are in Section 3.1 (baseline subtraction made explicit), Section 3.2 (facility-type restriction and fraction-destroyed recalibration), Section 3.3 (envelope distribution defined), and this section (version history added).

7 Worked Example: Ahvaz/Karoon, 2026-02-28

Context. As of 2026-05-24, the system has processed 47 fire events across 7 Iranian facilities under methodology v1.0.5. Of these, 27 events have FRP-based emission estimates; a subset also carry inventory-based estimates. Two events are fully reconciled ($\rho \in [0.5, 2.0]$); none are flagged as requiring review. The cumulative headline is $\sim$75,619 tCO_2e (p50) across all tracked facilities.

Event. The Ahvaz/Karoon production area event of 2026-02-28 is the largest single event in the current dataset. Its inputs are:

- Peak FRP: 114 MW.
- Duration: 6.5 days.
- Raw FRP integral: $I_{\text{raw}} = 2.893 \times 10^7$ MJ ($\approx$ 28,930 GJ), from VIIRS and MODIS overpasses.
- Pre-war baseline: $P_{75} = 13.59$ MW (from 12 months of archival observations at this facility).
- Destroyed-fraction assessment: triangular(0.05, 0.15, 0.30) (v1.0.5 default; no reviewer-assessed $\hat{\psi}$ available).
- Confidence label: REPORTED (satellite-persistent but no cloud-free optical confirmation obtained for this event window).

FRP-based point estimate. The following uses central prior values ($k_{\text{ext}} = 1.0$, $d = 0.70$, $\alpha = 0.368$ kg/MJ) to illustrate the equation chain before Monte Carlo sampling.
Stage 2 — Baseline subtraction (Eq. (4)):

$$\begin{aligned} I_{\text{net}} &= I_{\text{raw}} - \text{FRP}_{\text{baseline}} \cdot \Delta t \\ &= 2.893 \times 10^7 - 13.59 \text{ MW} \times 6.5 \text{ d} \times 86{,}400 \text{ s d}^{-1} \\ &= 2.893 \times 10^7 - 7.634 \times 10^6 \\ &= 2.130 \times 10^7 \text{ MJ}. \end{aligned}$$

Stage 3 — Overpass-gap correction (Eq. (5)):

$$I_{\text{FRE}} = k_{\text{ext}} \cdot d \cdot I_{\text{net}} = 1.0 \times 0.70 \times 2.130 \times 10^7 = 1.491 \times 10^7 \text{ MJ}.$$

Stage 4 — Conversion to CO_2 *mass* (Eqs. (6) and (7)):

$$\begin{aligned} m_{\text{fuel}} &= 0.368 \times 1.491 \times 10^7 = 5{,}487 \text{ t fuel}, \\ m^{\text{frp}}_{\text{CO}_2} &= 5{,}487 \times 3.15 = 17{,}284 \text{ tCO}_2\text{e}. \end{aligned}$$

Monte Carlo result. The point estimate (17,284 tCO_2e) falls between the fifth percentile and median of the full Monte Carlo distribution, as expected: the right skew in d and k_{ext} pulls the median above the product of central values. The realised distribution from the live system is:

p5	p50	p95
10,679 tCO_2e	22,704 tCO_2e	55,788 tCO_2e

This event is classified as REPORTED rather than VERIFIED or CONFIRMED because no cloud-free Sentinel-2 scene was obtained for the 28 February event window. Its contribution to the headline dashboard total is included, but its higher confidence threshold will be revisited if optical imagery becomes available.

References

[1] U.S. Environmental Protection Agency. *AP-42: Compilation of Air Emissions Factors*, Section 1.3 (fuel oil combustion). EPA.

[2] IPCC (2006). *2006 IPCC Guidelines for National Greenhouse Gas Inventories*, Volume 2, Chapter 1, Table 1.3. IPCC.

[3] Wooster, M.J., Roberts, G., Perry, G.L.W., Kaufman, Y.J. (2005). Retrieval of biomass combustion rates and totals from fire radiative power observations. *Journal of Geophysical Research*, 110, D24311.

[4] Hobbs, P.V., Radke, L.F. (1992). Airborne studies of the smoke from the Kuwait oil fires. *Science*, 256(5059), 987–991.

[5] Climate and Community Institute (2026). *Iran 14-day conflict emissions brief*. CCI.

[6] Conflict and Environment Observatory (2026). Incident records, Iran conflict. `https://ceobs.org`.

[7] Elvidge, C.D., Zhizhin, M., Hsu, F.-C., Baugh, K. (2016). Methods for global survey of natural gas flaring from Visible Infrared Imaging Radiometer Suite data. *Energies*, 9(1), 14.

[8] Freeborn, P.H., Wooster, M.J., Roberts, G. (2014). Addressing the spatiotemporal sampling design of MODIS to provide estimates of the fire radiative energy emitted from Africa. *Remote Sensing of Environment*, 140, 697–734.

[9] Neimark, B., Otu-Larbi, F., Bigger, P. (2026). Carbon emissions from the Gaza war. *One Earth*.

[10] De Klerk, L., Shlapak, M., Krakovska, S. et al. (2024). *Guidance on the Assessment of Conflict-Related GHG Emissions, v1.0.* Initiative on GHG Accounting of War / Ecoaction.

[11] Bigger, P., Neimark, B., Otu-Larbi, F. (2026). Two weeks of war in Iran unleashed more carbon pollution than Iceland. Climate and Community Institute Research Snapshot.

[12] Cottrell, L. (2022). A framework for military greenhouse gas emissions reporting. Conflict and Environment Observatory (CEOBS).

[13] Van Den Hoek, J., Scher, C. (2024). Sentinel-1 SAR coherent change detection for conflict damage assessment. Oregon State University Conflict Ecology Lab.

[14] Crawford, N. (2019). *Pentagon Fuel Use, Climate Change, and the Costs of War*. Costs of War Project, Brown University.